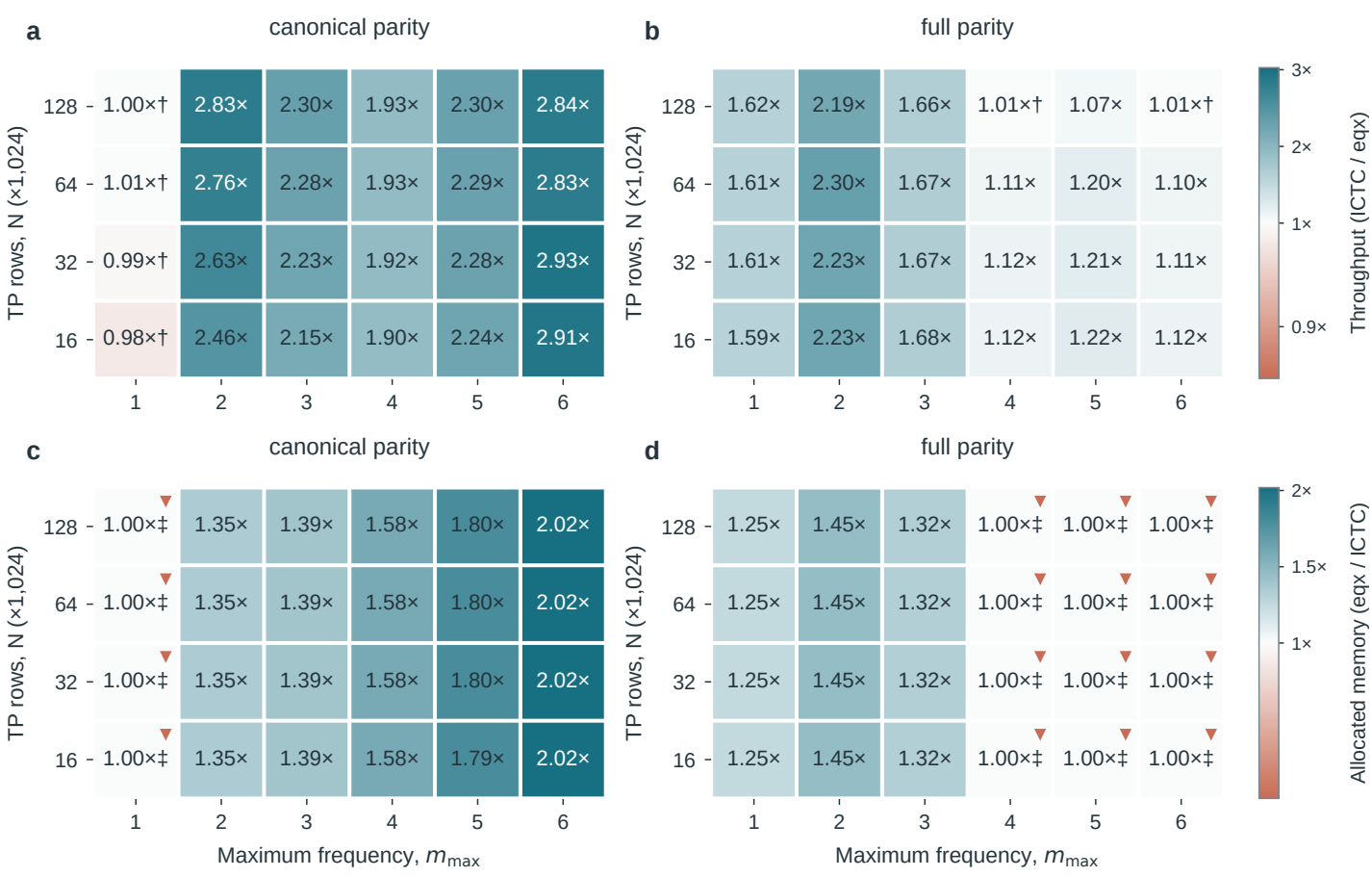

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · C = 256 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.00×† 2.83× 2.30× 1.93× 2.30× 2.84×
1.01×† 2.76× 2.28× 1.93× 2.29× 2.83×
0.99×† 2.63× 2.23× 1.92× 2.28× 2.93×
0.98×† 2.46× 2.15× 1.90× 2.24× 2.91×
1.62× 2.19× 1.66× 1.01×† 1.07× 1.01×†
1.61× 2.30× 1.67× 1.11× 1.20× 1.10×
1.61× 2.23× 1.67× 1.12× 1.21× 1.11×
1.59× 2.23× 1.68× 1.12× 1.22× 1.12×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1 2 3 4 5 6
Maximum frequency, $m_{\max}$
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5